SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID –SIH25126**
- **Problem Statement Title-CROPSATHI**
- **Theme-Agriculture**
- **PS Category- Software**
- **Team ID-MITADTSW073**
- **Team Name -Agrointel**

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CROPSATHI: AI-Powered Crop Price Forecasting & Advisory App

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Solution: Mobile-first platform for smallholder farmers with AI/ML-driven crop price forecasts, hyperlocal advisories, and community support.

Features:

📈 AI-based price forecasting & mandi comparisons
🌦️ Hyperlocal weather & smart advisories
🗣️ Multilingual + voice-first, pictorial, offline-ready app
👩‍🌾 Farmer groups, collective bargaining & knowledge sharing
🏛️ Govt. scheme & digital payment integration

Problem → Solution:

🔔📈Price uncertainty → Predictive alerts & advisories
🎤🗣️Low literacy → Voice + regional languages
📶❌📲Poor internet → Offline-first design
🔗📊Fragmented info → Single integrated platform
👥🤝Weak bargaining → Community marketplace

Innovation & Uniquene:

🤖📊 AI-driven price forecasting + advisory in one app
🗣️🌍 Multilingual & voice-first for inclusivity
📶🚜 Offline-first for rural connectivity
📍🌦️ Hyperlocal, personalized advisories
🏛️💰 Integrated govt schemes & payments
👩‍🌾👨‍🌾🤝 Farmer community empowerment

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TECHNICAL APPROACH

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👨‍💻 Programming & Frameworks

- 🐍 Python (FastAPI/Flask, scikit-learn, TensorFlow, XGBoost, LSTM)
- 📱 Flutter (Cross-platform Mobile App)
- 🌐 Node.js (Auxiliary APIs)

🗄️ **Database**

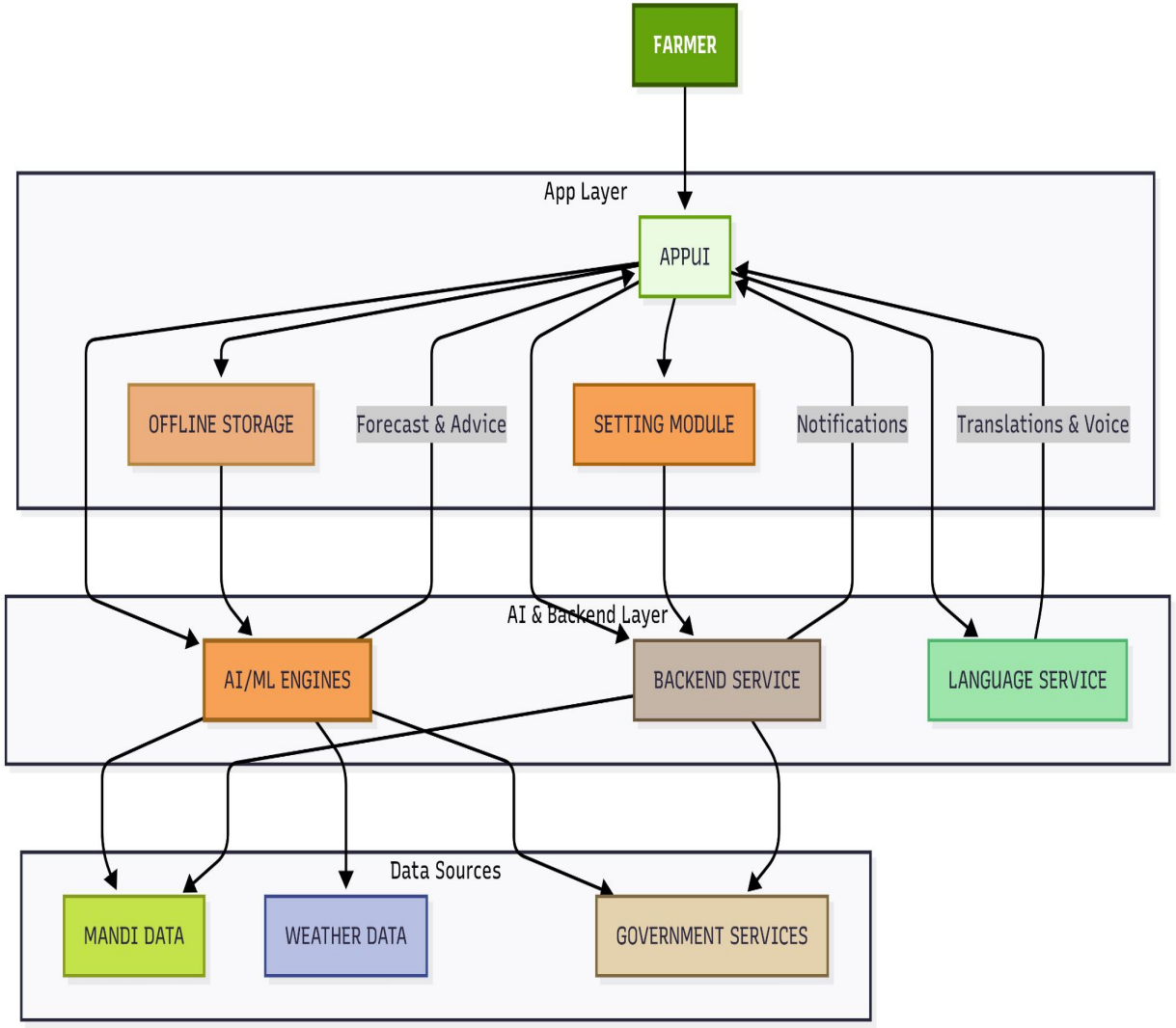
- 🐘 PostgreSQL + PostGIS (Geo-spatial support)

🔗 APIs & Data Sources

- 📊 Agmarknet + State Mandi APIs (Price data)
- 🌦️ IMD / OpenWeatherMap (Weather & Forecasts)
- 🛰️ Satellite APIs (NDVI/EVI – ISRO Bhuvan, Sentinel)
- 📲 SMS/USSD/WhatsApp APIs (Farmer alerts)

☁️ Cloud, Security & DevOps

- ☁️ NIC/MeghRaj Cloud + Edge Computing
- 🔒 AES, SHA-256, HTTPS, OTP Login
- 🐳 Docker + CI/CD Pipelines



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FEASIBILITY AND VIABILITY

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FEASIBILITY

Multi-platform access: app 📱 , USSD, IVR, kiosks , radio

Supports English, Hindi, Marathi, Gujarati 🗣️

Scalable to pest alerts , insurance , soil health 🌍

Personalized crop , price , and advisory alerts 🔔

Integration-ready with govt. APIs & FPOs

VIABILITY

Real-time price forecasts & mandi trends 📈

Outreach via kiosks , radio , WhatsApp , FPOs

Trusted data validated by KVKs & govt. bodies 🏛️

Impact dashboards for farmers 👨‍🌾 & policymakers

Nationwide scalability across states 🌏

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IMPACT AND BENEFITS

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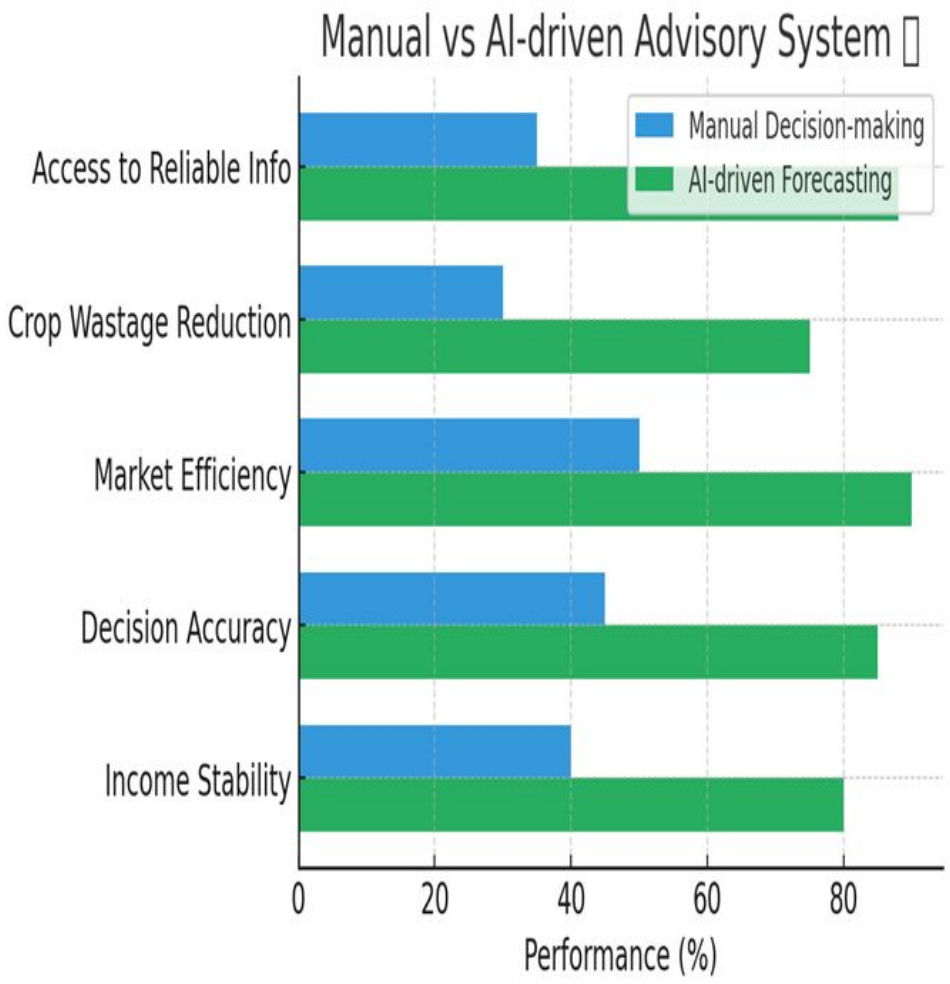
🌱IMPACT

💪 Increases farmers' bargaining power, reduces distress sales
📊 Enables smarter decisions with timely price forecasts
📱 Provides inclusive access via smartphones, USSD/IVR, kiosks, and radio
🤝 Builds trust through validation by KVKs, FPOs, and feedback loops

🌟 BENEFITS

💰 10–15% higher income through better sale timing
📉 Saves input costs with data-driven planning
⚖️ Improves market efficiency and ensures fair pricing
🌿 Promotes sustainable cropping patterns and reduces food wastage
🧑‍🎓 Enhances farmer dignity, independence, and digital literacy in local languages
🌍 Builds climate resilience with timely alerts



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RESEARCH AND REFERENCES

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- Enhancing agricultural commodity price forecasting with deep learning
  https://www.nature.com/articles/s41598-025-05103-z
  Demonstrates superior accuracy of LSTM/deep learning methods in crop price prediction; basis for our AI approach.
- Predicting Prices of Staple Crops Using Machine Learning
  https://www.mdpi.com/2071-1050/17/12/5456
  Compares multiple ML algorithms for agricultural price prediction; informs our choice of hybrid models.
- Crop Price Estimation Using Stacking Ensemble Technique
  https://ijisae.org/index.php/IJISAE/article/download/5801/4546/10878
  Reviews ensemble and stacking models that enhance forecast reliability; supports our meta-ensemble strategy.
- An Innovative Deep Learning Based Approach for Accurate Agricultural Crop Price Prediction
  https://arxiv.org/abs/2304.09761
  Explores deep learning and remote sensing integration for price forecasting; justifies our NDVI/EVI satellite features.
- Market Intelligence in India Price Linkages and Forecasts
  https://ageconsearch.umn.edu/record/344987/files/Market%20Intelligence%20in%20India.pdf
  Provides insight into Indian market price linkages and forecasting; context for our app's real-world relevance.

IMPORTANT INSTRUCTIONS

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Please ensure below pointers are met while submitting the Idea PPT:

1. **Kindly keep the maximum slides limit up to six (6). (Including the title slide)**
2. **Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures**
3. **Keep your explanation precise and easy to understand**
4. **Idea should be unique and novel.**
5. **You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).**
6. **You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.**

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.